SDG-BASED DATA VISUALIZATION PROJECTS

Using Pandas and Matplotlib

1. District-wise Health Indicators Dashboard – SDG 3

Aim

To develop a dashboard using Pandas and Matplotlib to monitor and compare district-wise health indicators, identify healthcare gaps, and support evidence-based decision-making under Sustainable Development Goal 3 (Good Health and Well-being).

Objectives

- Collect and organize district-wise healthcare data using Pandas.
- Calculate and compare important health indicators across districts.
- Create clear visualizations using Matplotlib.
- Identify districts with relatively high-risk health indicators.
- Present the results in a simple dashboard for monitoring and analysis.

Suggested Dataset Attributes

Attribute	Description	Unit
District	District name	Text
State	State name	Text
Infant Mortality Rate	Infant deaths per 1,000 live births	Per 1,000
Maternal Mortality Rate	Maternal deaths related to childbirth	Rate
Immunization Rate	Children receiving required immunization	%
Life Expectancy	Average expected lifespan	Years
Hospital Beds	Hospital beds available	Per 1,000
Doctor Availability	Doctors available to population	Per 1,000
Institutional Delivery	Deliveries conducted in health institutions	%

Visualization Techniques

- Bar Chart – Compare health indicators across districts.
- Line Chart – Analyze health-indicator trends over time.
- Pie Chart – Show the distribution of healthcare coverage.
- Scatter Plot – Examine the relationship between healthcare availability and mortality.
- Heatmap – Compare multiple health indicators across districts using a compact matrix.

Python Implementation

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("health_indicators.csv")

# Display dataset
print(df.head())

# District-wise comparison
district_data = df.groupby("District")["Immunization Rate"].mean()

plt.figure(figsize=(12, 6))
district_data.sort_values(ascending=False).plot(kind="bar")

plt.title("District-wise Immunization Rate")
plt.xlabel("District")
plt.ylabel("Immunization Rate (%)")
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()

# Mortality vs healthcare availability
plt.figure(figsize=(8, 6))
plt.scatter(df["Hospital Beds"], df["Infant Mortality Rate"])
```

```
plt.title("Healthcare Availability vs Infant Mortality")
plt.xlabel("Hospital Beds per 1,000")
plt.ylabel("Infant Mortality Rate")
plt.tight_layout()
plt.show()
```

Expected Outcome

The dashboard will provide a visual overview of district-level health conditions. It can help identify districts requiring greater healthcare attention, compare immunization and healthcare infrastructure, and understand relationships between healthcare availability and mortality. The project directly supports SDG 3.

2. Comparative Analysis of Renewable Energy Adoption Across Indian States – SDG 7 & 13

Aim

To conduct a comparative analysis of renewable energy adoption across Indian states using Pandas and visualization techniques, with emphasis on clean energy adoption and climate action.

Objectives

- Compare renewable energy capacity among Indian states.
- Analyze the contribution of solar, wind, hydro and biomass energy.
- Study renewable energy growth over different years.
- Identify states with high and low renewable energy adoption.
- Use visual analytics to support SDG 7 and SDG 13 objectives.

Suggested Dataset Attributes

Attribute	Description	Unit
State	Indian state name	Text
Year	Year of observation	Year
Solar Capacity	Installed solar power capacity	MW
Wind Capacity	Installed wind power capacity	MW
Hydro Capacity	Installed hydro power capacity	MW
Biomass Capacity	Installed biomass capacity	MW
Total Renewable Capacity	Combined renewable capacity	MW
Total Power Capacity	Total installed power capacity	MW
Renewable Energy Share	Share of renewable capacity	%

Visualization Techniques

- Bar Chart – Rank Indian states by total renewable energy capacity.
- Stacked Bar Chart – Compare solar, wind, hydro and biomass contributions.
- Line Chart – Show renewable energy growth over the years.
- Pie Chart – Display the overall contribution of renewable energy sources.
- Scatter Plot – Analyze the relationship between renewable capacity and total power capacity.

Python Implementation

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("renewable_energy_india.csv")

# Display dataset
print(df.head())

# State-wise renewable capacity
state_data = df.groupby("State")["Total Renewable Capacity (MW)"].sum()

plt.figure(figsize=(12, 6))
state_data.sort_values(ascending=False).plot(kind="bar")

plt.title("Renewable Energy Adoption Across Indian States")
plt.xlabel("State")
plt.ylabel("Renewable Capacity (MW)")
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()

# Renewable sources comparison
source_data = df[["Solar Capacity (MW)",
               "Wind Capacity (MW)",
               "Hydro Capacity (MW)",
               "Biomass Capacity (MW)"]].sum()

plt.figure(figsize=(8, 6))
source_data.plot(kind="pie", autopct="%1.1f%%")
plt.title("Renewable Energy Source Distribution")
```

```
plt.ylabel("")
plt.show()
```

Expected Outcome

The analysis will show differences in renewable energy adoption among Indian states, highlight the contribution of different renewable sources, and reveal growth patterns. The findings support SDG 7 (Affordable and Clean Energy) by encouraging renewable-energy adoption and SDG 13 (Climate Action) by supporting the transition toward lower-carbon energy systems.

Conclusion

Pandas and Matplotlib provide an effective approach for transforming SDG-related datasets into meaningful visual information. The two projects demonstrate how data analysis and visualization can be used to monitor health outcomes and evaluate renewable-energy adoption. Such dashboards can support monitoring, comparison, and data-driven planning.